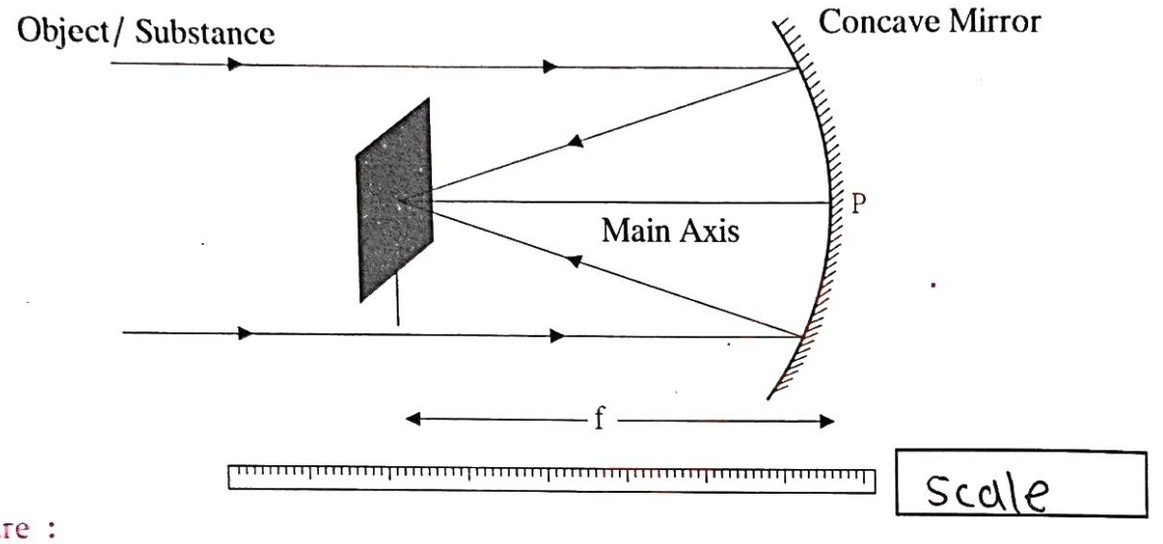
Experiment No. 9

Title : To determine the focal length of concave mirror by obtaining the images of an object at different distances.

Apparatus : Concave mirror, Stand to hold the mirror, Meter scale, Screen on stand etc.

Figure :



Procedure :

1. Fit a concave mirror on stand properly.
2. Take an object and fit it on a stand.
3. Fit a screen on a stand and keep it before the mirror.
4. Keep the object before the mirror and measure the distance between the mirror and object.
5. Obtain a sharp image of the object on screen to and fro.
6. Measure the distance between the mirror and the screen.
7. Change the distance of the object before the mirror and repeat the steps 4 to 6 in the procedure.

Observation : **Least count of the meter scale = ...mm**

Sr. No.	Distance between mirror and object (u)	Distance between mirror and screen (v)	$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	Focal Length f (cm)
1	- 30	-15	$\frac{1}{-15} + \frac{1}{-30} = \frac{1}{F} = \frac{1}{-10}$	10
2	-10	5	$\frac{5}{10} + \frac{1}{-10} = \frac{1}{F} = \frac{1}{10}$.	10
			Average Focal Length = -5	

Inference : Focal length of the concave mirror = ...cm.

.....when length falls a body part of it is reflected part plane mirror convexe mirror is the three mirror are use in verious equipment.

Multiple Choice Questions

1. The image of a distant object is obtained at.....of the concave mirror.
   A. Focus (f) B. Centre of Curvature (C) C. 2 f D. 2 v
2. Virtual image is obtained when the object is placed ..... with refrence to concave mirror.
   A. at infinite distance B. at the focus
   C. at the centre of curvature D. between focus and pole
3. Which mirror is of different type?
   A. Denitist's mirror B. Mirror used in vehicle headlight
   C. Rear view mirror of car D. Solar cooker mirror
4. Which of the following property is of real image?
   A. Inverted B. Can be taken on screen
   C. Erected D. Can not be taken on screen

Questions

1. Write various uses of concave mirror in everyday life.

Barber shop dental hospital :: If the object is placed between the pole and focus of the mirror a react virtual and magnified image is objectend mirror an both side of cars are convexe mirror are fitted is the shops torchi lamp etc.

2. How is a water heater made with the help of a concave mirror?

various equipment using salar system such very reflected by a concave mirror or come to ge ther in focal length the all length towards the light place pole are taken now positive while these measured vertically towards principle axis is negative therefore convex is (x).

3. What will happen if a convex mirror is used in the experiment?

The nature of image formed by a convex mirror does not depend on the distance of the object from the mirror these image are always victural smaller than the object and the mirror.

VREANS